What Does Deferral Buy? Measuring Provisional Identity in Streaming Linkage [Experiment, Analysis & Benchmark]

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ABSTRACT

A streaming linker can defer an uncertain report's final identity assignment while keeping it retrievable. We evaluate this choice using an event-time engine that emits one irrevocable outcome per request, enforces a seven-day deadline, and records complete outputs. On San Diego's May stream, five matched scorer fits give mean pairwise F_1 of 0.415 for arrival commitment, 0.449 for a score band, and 0.467 for a staleness gate. The gate exceeds a ten-day cutoff by 0.0246 (95% fit-seed interval $[0.0093, 0.0400]$). A retrospective July evaluation under the newly frozen protocol gives a gate–cutoff difference of 0.0180 ($[0.0135, 0.0225]$). Hiding pending identities changes both retrieval and realized waiting. Event-level analyses track newly recovered and lost positive pairs, changes in merge denominators, and claims initiated both at arrival and at a deadline. Graphs built from actual candidates distinguish direct coverage from transitive reachability. The logistic scorer reverses the gate–cutoff ordering on July. Together, these measurements characterize the quality and timing of deferred linkage against workflow-derived identity labels.

KEYWORDS

entity resolution, streaming, decision timing, prefix-safety

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1 INTRODUCTION

A municipal service desk may receive several reports about a broken streetlight. Linking them can avoid duplicate work, while an incorrect link can suppress an unrelated problem. If linkage triggers an irreversible action, the system must decide when a tentative identity becomes binding. A record can remain unresolved while still being visible to later reports (Figure 1). This changes what later records retrieve and what evidence is available when the original record is finally assigned.

An immediate policy binds an identity at arrival. A causal policy can bind it later, using only information available at emission. Delayed decisions and irreversible matching already have a substantial history in online matching and streaming systems [1, 12]. We ask what a learned streaming linker gains from a retrievable provisional identity, and how much of any gain a simple age cutoff can reproduce.

The measured system choice. Our pipeline retrieves candidate issues using address and token indexes and scores their representatives. Arrival commitment either creates an issue or merges immediately. A score band instead holds marginal matches; a gate holds only marginal matches to sufficiently old representatives. Each held record forms a singleton provisional identity with a timer. A later merge can finalize that target before attaching to it, or the timer can re-score it. Previously committed identities never change. This protocol lets us compare state management while keeping the scorer and retrieval budgets fixed.

Realized candidate sets depend on the commitment policy as well as the retrieval settings. A provisional identity may become a candidate for later requests and may change the representatives and capacities they encounter. We therefore record actual candidate snapshots and distinguish end-to-end policy comparisons from a conditional analysis of one frozen candidate trace. The frozen analysis conditions on the recorded trace, while adaptive replay captures changes in future candidates.

Evidence from matched comparisons. On May, the score band changes mean F_1 from 0.415 to 0.449, and the gate reaches 0.467. The ten-day cutoff reaches 0.442. The gate–cutoff contrast is 0.0246, with a five-fit interval of $[0.0093, 0.0400]$ (Table 1). May had influenced earlier exploration, so we retain its exploratory status. We

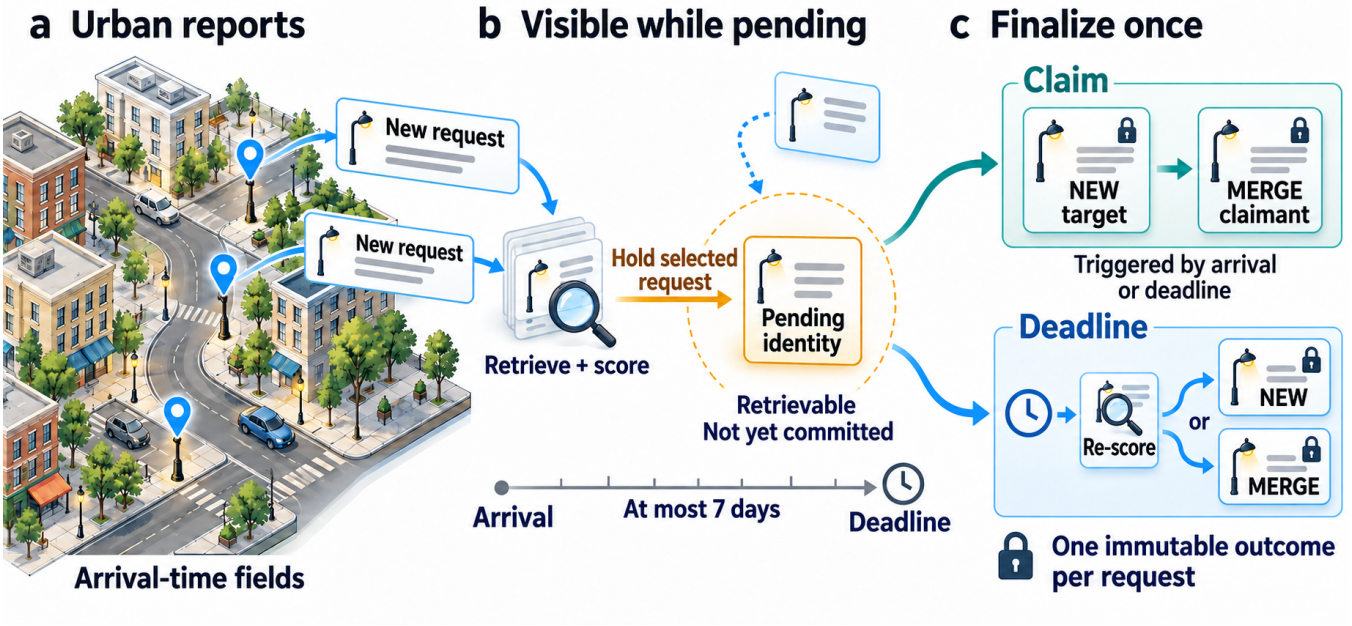


Figure 1: Selected deferred path. One request remains retrievable while pending. A claim emits the target's NEW before the claimant's MERGE and can originate at arrival or deadline. Otherwise, deadline re-scoring emits NEW or MERGE. Each request has one immutable outcome. AI-assisted schematic checked against Algorithm 1; immediate paths omitted.

freeze the complete revised protocol before evaluating July. On that stream, the same gate and cutoff configurations differ by 0.0180 ([0.0135, 0.0225]). These intervals quantify scorer-fit variation on fixed monthly streams.

Event-level analysis shows which pairs change and when requests are finalized. Visibility-off controls change early claim opportunities and actual waiting times as well as retrieval. Correct-pair sets contain both additions and losses, and claims can originate at an arrival or another record's deadline. A higher adjudicable-merge fraction can coexist with fewer adjudicable merges in absolute terms.

Contributions. We specify an irrevocable streaming-linkage protocol with explicit event timing and provide executable output-prefix checks. We compare matched policies and scorers with a recorded selection timeline, and analyze events and candidates to distinguish pair changes, merge-support changes, and finalization triggers. The accompanying artifact preserves inputs, configurations, result rows, and verification code.

2 RELATED WORK

Record linkage, online matching with delays, and streaming systems address uncertain matches, delayed decisions, and evolving results. We examine how selective commitment and the visibility of provisional identities interact in a particular streaming linkage pipeline.

Entity resolution and incremental maintenance. Duplicate detection addresses multiple representations of the same entity [11]. Swoosh formalizes match and merge functions and properties that permit efficient resolution [6]. Incremental ER reuses earlier materializations when matching rules or data change [36]. Gruenheid et al. also use new records to correct earlier linkage errors [17]. Our protocol fixes the scorer and preserves finalized assignments. It controls when a record becomes final and whether its provisional identity remains retrievable.

Progressive processing. Progressive ER prioritizes useful partial results under limited computation. Pay-as-you-go methods use hints to prioritize likely matches [37]. Relational methods adapt resolution plans as resolved relationships supply new evidence [2]. Schema-agnostic methods schedule comparisons over heterogeneous inputs [34]. Maciejewski et al. organize the design space into filtering, weighting, scheduling, and matching, and discuss incremental approaches to streams [26]. Progressive ER can therefore use evolving evidence. Our internal comparisons hold retrieval settings and the learned scorer constant while varying commitment and provisional visibility. Policies alter index state and therefore the candidate pairs realized under these fixed settings.

Matching models and workflows. Magellan supports the entity-matching workflow, including exploration, labeling, and debugging [21]. DeepMatcher explores neural matching architectures [30], while Ditto fine-tunes pretrained language models for pair classification [23]. These works address workflow or representation choices. We hold the scorer fixed to compare commitment policies within the tested pipeline.

Online matching with delays. Emek, Kutten, and Wattenhofer study minimum-cost perfect matching with delays in a known metric space [12]. Requests arrive online, may wait to be paired with

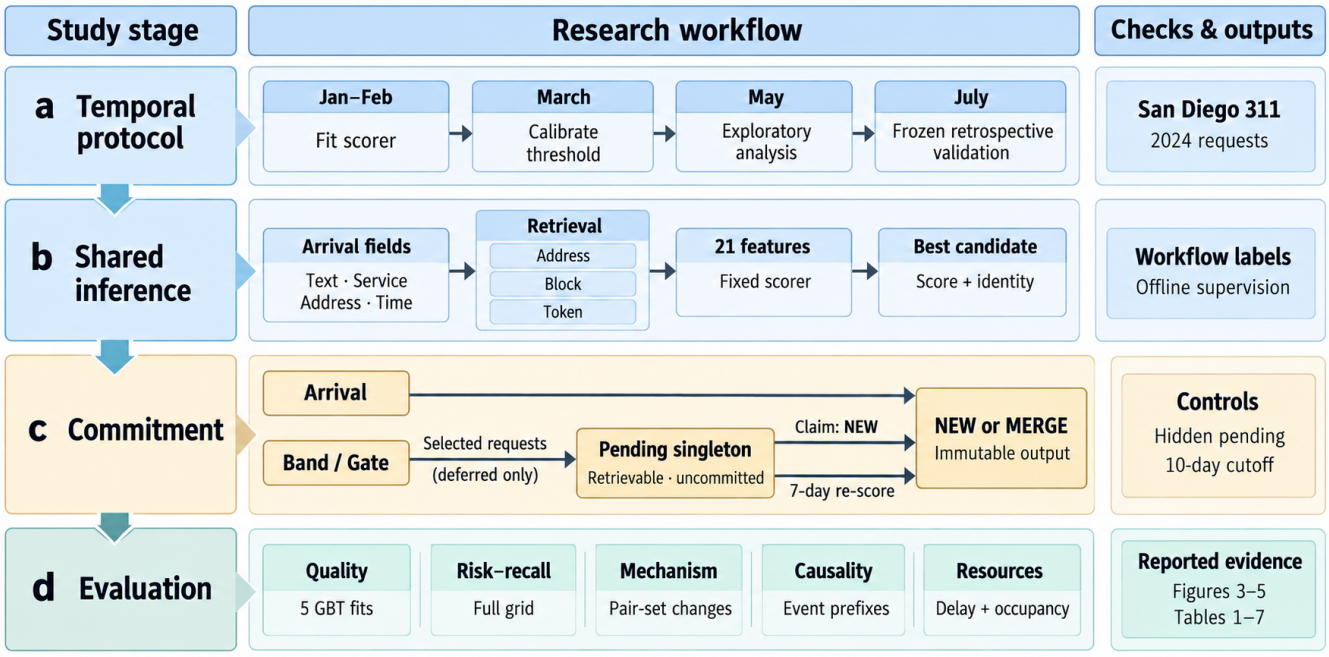


Figure 2: Study design. January–February fits the scorer; March calibrates its threshold by a complete grid. May remains exploratory and July evaluates the newly frozen configuration. Address, block, and token retrieval feed a binding event-time state machine. Workflow labels supervise fitting and evaluation, while future reference labels are never online lookup inputs. AI-assisted schematic checked against the released protocol and implementation.

later arrivals, and are matched without preemption. The objective combines matching distance and waiting time, with competitive guarantees relative to an offline optimum. Commitment timing and irreversibility are already central to this model. Our setting differs in its learned scores over heterogeneous records, multi-record issue states, and budgeted retrieval from a changing index. We evaluate linkage quality relative to workflow-derived partitions and measure logged merge operations and latency, rather than a metric matching cost. Our contribution is an empirical study of these system choices.

Abstention and deferral. The Fellegi–Sunter framework distinguishes links, non-links, and possible links requiring further review [13]. Reject-option classification trades coverage against error [7, 15]; learning to defer explicitly routes selected decisions to an expert [29]. Abstention and delegation are distinct design choices. In our implementation, an uncertain record remains available to the same linking system, which may act on subsequent arrivals or re-score it when its waiting interval expires. Comparing visible and invisible pending records tests this state-management choice within the evaluated linker.

Streaming completeness and output revision. The Dataflow model separates event-time windows from triggers that determine when results are emitted, and supports different refinement modes, including accumulation and retraction [1]. Watermarks provide progress estimates; heuristic watermarks do not certify that all relevant data have arrived. This literature exposes correctness–latency trade-offs. Our application contract requires emitted merge decisions to remain binding. We ask how to manage uncertain identity state before emission under that contract.

Candidate generation and external comparisons. Blocking and filtering reduce comparison cost while affecting which direct record pairs reach a matcher [8, 32]. DeepBlocker learns tuple representations for candidate generation [35]; Sparkly uses lexical top-k TF/IDF blocking [33]. Our address, block, and token retrieval is one choice of candidate-generation procedure. Direct candidate-pair recall and final partition recall are different quantities: transitive merges can recover a pair that was never directly compared. The reported candidate coverage characterizes the measured retrieval procedure. Our internal policies enforce a true size cap of 20. Splink uses its own blocking and estimation procedures [28]; its comparison therefore evaluates complete systems with their respective candidate populations.

Risk calibration under distribution shift. Conformal risk control provides risk guarantees under stated conditions [3]; adaptive and nonexchangeable conformal methods address settings beyond ordinary exchangeable calibration [5, 16]. Our threshold calibration and subsequent monthly evaluation are empirical procedures. They do not provide an online guarantee on the false-merge proportion. Our threshold-policy comparisons concern the evaluated settings and their observed risk–coverage trade-offs.

Evaluation and scope. Comparative ER studies document sensitivity to configuration and training data [22]; benchmark re-evaluation examines how task construction affects the difficulty

of learned matching [31]. ER and coreference measures assign different weights to record-level, pair-level, and entity-level errors [25, 27]. We report pairwise partition metrics together with decision-level merge errors to describe both final partitions and the sequence of emitted actions. These measures are complementary; record-averaged measures can also be sensitive to erroneous merges. Our conclusions concern the observed streams, their labeling procedures, and the tested configurations. Aggregate lag-bin changes and policy differences provide descriptive associations within these streams.

3 PROTOCOL AND MEASUREMENTS

Arrival, commitment, and identity. A request x_i has arrival time a_i, final commitment time c_i, and final issue identity d_i. Time is a UTC timestamp, distinct from the record index i and the score threshold τ. A final output event contains its emission time, request identifier, action, target identity, payload, and sequence number. New creates a stable identity; Merge attaches one request to an already finalized identity. Each observed request emits exactly one such event. An internal pending state is not a final output. A committed request is never moved to another identity, and finalized identities are never merged together.

Definition 1 (Immediate commitment). *An immediate policy satisfies $c_i = a_i$ for every request. Its action and identity use only the input prefix available at a_i and fixed prior state.*

Definition 2 (Bounded causal commitment). *For a fixed maximum delay D, a bounded causal policy satisfies $a_i \le c_i \le a_i + D$. The final identity d_i and its output event use only arrivals no later than c_i and fixed prior information. Conditional on the same initial state, scorer, configuration, and randomness, two streams identical through any time T must emit identical ordered event sequences through T, including event counts, times, objects, actions, targets, and payloads. The identity d_i is binding at c_i. Here D is an elapsed duration; a reported day is exactly 86,400 seconds.*

Immediate commitment is the special case $D = 0$. Nonanticipation allows evidence arriving after a_i but no later than c_i to influence a delayed output. These definitions concern inference with a fixed scorer. Historical training-label availability requires separate verification.

Workflow reference partition. We construct undirected components from the available official child–parent links. The labeled domain $\mathcal{K}$ contains records participating in these links. For two records in $\mathcal{K}$, their pair is positive if they belong to the same component and negative otherwise under a *closed-world workflow assumption*. Cross-component links may be missing. A pair touching an unlabeled record is unknown. Accordingly, "correct" and "wrong" below mean agreement or disagreement with this frozen workflow partition, rather than an independently established operational event identity. Attributing a counted error to recurrence or operational inconvenience requires additional labels.

Two different units of error. Partition precision, recall, and F_1 use all unordered predicted pairs in $\mathcal{K}$ and all positive reference pairs whose endpoints occur in the replay window. In contrast, a merge decision is the directed pair of the finalized request and the first representative of its chosen destination. Let C, W, and U count correct, wrong, and unknown merge decisions, and let $M = C + W + U$. We report

$$\mathrm{FMP}_{\mathrm{dec}} = \frac{W}{C+W}, \qquad \mathrm{coverage}_{\mathrm{dec}} = \frac{C+W}{M}.$$

Both ratios are undefined when their denominators vanish. Coverage is the adjudicable fraction of merges, not the fraction of requests receiving service. For every arm we also count total requests, final events, New, Merge, pending requests, and each of C, W, U. A single merge may induce several partition pairs; the two error units must not be interchanged.

4 AN IRREVOCABLE EVENT-TIME IMPLEMENTATION

Candidates and scores. The index has three channels: exact normalized address within service, hundred-block address within service, and token overlap. Each address channel selects a contiguous slice of up to 40 calendar-ordered postings around the query date's insertion point. This differs from selecting the nearest postings by absolute distance. At arrival, these slices contain recent entries; deadline rescoring retains the request's original date. The text channel accumulates inverse-document-frequency weights for the eight highest-IDF query tokens, scanning the latest 60 postings per token and retaining up to 60 identities. The deduplicated union can contain up to 140 identities. Candidate identities are deduplicated and ordered by their stable identity keys. We score the first representative of each eligible identity. The index is pruned on UTC day changes. At each scoring event, an additional exact elapsed-time check excludes identities whose latest member is more than 60 days old; self-candidates and identities with 20 actual members are also excluded. A cutoff arm additionally excludes representatives older than its specified cutoff at scoring time.

Our boosted classifier receives 21 feature slots describing lexical/address, service, time, and issue state. In the singleton-candidate training replay, issue size is constant at $\log(2)/\log(11)$, the historically named `service_rare` slot is a constant nonempty-service indicator, and both unused address-embedding slots are zero. The artifact lists every feature, its transform, and its observed training range. Score ties use the lexicographically smallest stable identity key, and token-IDF ties use lexical token order. Index postings are added once, when an identity first becomes visible; they are not rebuilt after every attached member.

Policies. Let s^* be the best candidate score. Arrival commitment merges when $s^* \ge \tau$ and otherwise creates a new identity. The band policy holds a request when $D > 0$ and $\tau \le s^* < \tau+\delta$. The gate holds it only when this band condition holds and the best representative is at least h elapsed days old. Otherwise it follows arrival commitment. A pending identity contains exactly one uncommitted request. Its timer is scheduled for $a_i + D$; by default it is immediately retrievable. Visibility-off policies retain the same hold predicate but hide this identity until it is finalized as New.

Claims and timers. A later request claims a pending target only when its own policy selects an actual merge. The target first emits New, then the claimant emits Merge into that now-final identity; the target timer is canceled. If the later request is itself held, it

Algorithm 1 Bounded causal band or gate policy

Require: Fixed scorer, τ, δ, D, h, visibility v, capacity $B \geq 1$

procedure FinalizeNew(p, t, g)
  Assert p is a singleton with an uncommitted request
  Cancel its timer; remove pending status; ensure p is visible
  Emit final New for its request at t, trigger g
end procedure
procedure Attach(x, p, t, s, g)
  Assert x is uncommitted, $x \notin p$, and $|p| + 1 \leq B$
  if p is pending **then**
    FinalizeNew(p, t, claim)
  end if
  Cancel and retire x's pending singleton, if present
  Attach only x; retain the latest member date if newer
  Emit final Merge(x, p) at t, score s, trigger g
end procedure
procedure Arrive(x, t)
  Process all timers due at or before t
  $(p, s) \leftarrow$ BestEligible(x, t)
  if $p = \emptyset$ or $s < \tau$ **then**
    Create stable singleton p_x; FinalizeNew(p_x, t, arrival)
  else if $D > 0$, $\tau \leq s < \tau + \delta$, and (band or age$(p, t) \geq h$) **then**
    Create singleton pending p_x; schedule $t + D$
    if v **then**
      Index p_x
    end if
  else
    Attach(x, p, t, s, arrival)
  end if
end procedure
procedure Deadline(x, t)
  if x is already final **then**
    return
  end if
  $p_x \leftarrow x$'s pending singleton
  $(p, s) \leftarrow$ BestEligible(x, t), excluding its own identity
  if $p \neq \emptyset$ and $s \geq \tau$ **then**
    Attach(x, p, t, s, deadline)
  else
    FinalizeNew(p_x, t, deadline)
  end if
end procedure
procedure Finish
  Process all remaining timers at their individual exact due times
end procedure

creates a separate singleton pending identity. A deadline resolution can likewise claim another pending identity. Previously committed members never move to a different issue. Timers at a timestamp execute before arrivals at that timestamp, with stable identity order for tied timers and stable record order for tied arrivals. Identities use the key `I:` followed by their anchoring request identifier.

Lemma 1 (Finality and bounded causal output). *Given a fixed deterministic scorer, configuration, and initial state, and processing arrivals and timers in the stated order, Algorithm 1 emits one final event per observed request, never changes its final identity, and satisfies bounded causal commitment when the event clock is advanced through each deadline, including the final drain.*

Proof. Each transition reads only the current index and already-arrived records. Identical input prefixes therefore give identical state transitions, timer firings, and ordered outputs by induction. A request emits immediately, on a claim after its arrival, or at its scheduled deadline. Timers are processed at their exact due times, so no request remains unresolved after $a_i + D$. Every finalization removes the pending state, and stale timer entries are ignored. The attach operation accepts only an uncommitted source and first finalizes a pending target, so a final identity never changes and a request cannot emit twice. The capacity test counts actual members before attachment. □

This argument concerns the stated algorithm. Executable tests and the real-stream output audit below check the released implementation. At month end, we drain remaining timers at their exact due times, using only records supplied in that month's replay. We measure latency as $(c_i - a_i)/86400$ in days and separately summarize all records and held records. Entry into pending, claim finalization, and deadline resolution are distinct counters. Resource measurements include peak pending state and processing time under the reported local CPU conditions.

5 EXPERIMENTAL DESIGN

Data and retrospective labels. We use San Diego's 2024 closed-request snapshot from Get It Done [10]. The archived export contains 429,924 requests and 57,463 parent-link fields, of which 56,153 resolve within the snapshot. About 20.3% of requests participate in the linked domain. The closed-request sampling frame can underrepresent unresolved cases; the final export leaves post-arrival edits to description and address fields unverified. Scoring uses only the designated request fields, excluding closure/status fields. We build the reference components from the complete snapshot, including links that may have become available after the training period. Link-creation times are unavailable. The labels therefore provide retrospective weak supervision. Chronological feature replay and a fixed-model causal audit address inference-time behavior. Historical label availability and prospective deployment fidelity remain unverified. Legitimate training information and train–test separation are broader leakage concerns [19, 20] than the fixed-model output-prefix property tested here.

We select monthly windows by the source's local calendar, interpret its naive timestamps as America/Los_Angeles, and convert them to UTC before replay. Training and scoring use UTC calendar-day features; timers use exact elapsed seconds, including across daylight-saving changes. Each monthly replay starts with an empty index. The ordering key is UTC arrival time followed by the source request identifier. The artifact records this convention and the raw-file checksum.

Training. January–February contains 71,923 records. A fixed candidate replay uses the first 22,000, retains each weak negative candidate with probability 0.35 using sampling seed 13, and produces 441,455 training pairs, including 3,080 positives. Equal known components are positive; all other sampled pairs, including unknown ones, are treated as negative for fitting. This noisy-label convention differs from evaluation: pairs outside the known domain remain unadjudicated. All policies share these training pairs. The scorer uses stochastic gradient boosting [14] with 160 trees, learning rate 0.1, depth 3, subsample 0.5, and minimum leaf size 20. We refit it with seeds $\{1, 2, 3, 4, 13\}$. The second family is standardized logistic regression with $C = 1$, LBFGS, maximum 4,000 iterations, and seed 13.

Nonmonotonic calibration. For each family we run the arrival policy on March with seed 13 at thresholds .01, .02, .0358, .05, .07, .1, .15, .2, .3, .5, .7, .9, and .99. A point is volume-eligible only if it merges at least 1% of requests and has at least 100 adjudicable merges. Among volume-eligible points with decision FMP at most 0.40, we maximize positive-pair recall, breaking ties by higher F_1 and then higher τ. If none is feasible, the rule selects the minimum observed FMP among volume-eligible points, breaking ties by higher recall, F_1, and then τ. In that case, it also reports infeasibility. This finite-grid rule allows nonmonotonic responses: changing a threshold changes later identity states. The volume criterion excludes zero-merge runs. The 0.40 budget is a chosen workflow-proxy constraint, not a production safety guarantee.

Selection record and evaluation windows. Historical band, gate, and cutoff choices were influenced by experiments already examined. The old April cutoff sweep had five seeds for $X = 5, 10$ but only three for $X = 14$; its selection script excluded the incomplete candidate. The choice of $X = 10$ is conditional on this incomplete April sweep. Original first-view and freeze times cannot be recovered and remain marked unknown in the released stage manifest. No file modification time is presented as a preregistration timestamp.

For the repaired implementation we froze the full experiment protocol before viewing its outcomes on 27 October 2026. We retain $\delta = .5$, $D = 7$ days, $h = 5$ days, and cutoff $X = 10$ from historical exploration. March calibrates the new model-specific threshold; that choice is recorded before any revised May/July quality run. May remains exploratory. A metadata search found no earlier July quality outputs, so July supplies a newly frozen retrospective validation of the gate–cutoff comparison. It is not prospective deployment or a reconstruction of an earlier preregistration. All five fitted scorers use the same family-specific threshold and the same policy settings.

Controls and uncertainty. For May and July we replay arrival, band, gate, cutoff, invisible band, and invisible gate with all five shared seeds. The logistic check includes arrival, band, gate, and the same ten-day cutoff in both months. We retain the entire 13-point May boosted threshold curve, one-seed April parameter grids, and $D = 3, 14$ sensitivity runs. The main $D = 7$ rows complete that delay comparison. These secondary runs do not select the primary July comparison. Means and sample SDs describe five scorer fits on a fixed stream. Paired differences use exactly the same five seeds, with 95% t intervals (four degrees of freedom). Intervals are pointwise and unadjusted for multiple contrasts. They quantify fit-seed variability with the city streams, label acquisition, and historical policy-selection process held fixed.

Version separation. The historical engine could move previously committed members. It processed nominal deadlines only at later arrivals or at window termination. It also double-counted the first member in its issue-size counter. We therefore retrain all scorers after fixing the counter and report the repaired implementation as a separate experiment series. Primary tables use only this series. Historical transfer and external-linker results, where retained, are explicitly labeled as measurements of their historical implementations.

6 RESULTS OF THE CORRECTED REPLAY

6.1 Matched quality and scorer dependence

The gate improves on the ten-day cutoff in every matched boosted fit in both months (Table 1 and Fig. 3). Its mean difference is 0.0246 on May, or 2.46 percentage points of F_1, and 0.0180 on July, or 1.80 points. The paired 95% intervals are $[0.0093, 0.0400]$ and $[0.0135, 0.0225]$. The band–cutoff intervals include zero: May 0.0067 $[-0.0128, 0.0261]$ and July 0.0045 $[-0.0002, 0.0093]$. The evidence of improvement beyond this cutoff is therefore stronger for the gate.

On May, hidden Band minus Arrival is 0.0007, with interval $[-0.0132, 0.0145]$. Hidden Gate minus Arrival is 0.0101, with interval $[-0.0024, 0.0227]$. These estimates are imprecise; equivalence remains untested. Visible minus hidden effects are 0.0336 $[0.0185, 0.0488]$ for Band and 0.0422 $[0.0323, 0.0520]$ for Gate. Visibility changes both candidates and claim times, coupling retrieval and realized waiting in this contrast.

On July, hidden Gate minus Arrival is 0.0058 with paired 95% interval $[0.0036, 0.0081]$; all five fit differences are positive. Visible Gate adds 0.0266 $[0.0212, 0.0321]$ over hidden Gate. The hidden contrast combines waiting with changes in selection, deadline rescoring, and the state encountered later.

The policy ordering changes with the scorer (Table 2). In the single-fit logistic checks, Band lowers F_1 in both months. Gate exceeds Cutoff on May by 0.0253, but falls below it on July by 0.0077. A common numerical band width can represent different uncertainty levels across differently calibrated scorers.

6.2 Calibration and threshold alternatives

The March rule selects $\tau = .0358$ for boosting (decision FMP 0.3986, recall 0.4887) and .20 for logistic regression (FMP 0.3891, recall 0.4454). The selected boosted threshold exceeds the budget on May (FMP 0.4362). In the complete May grid, .15 satisfies the volume rule and has FMP 0.3979, recall 0.3944, and $F_1 = 0.4147$. Threshold adjustment is therefore a viable alternative on this observed grid. FMP is nonmonotonic: raising τ from .5 to .7 changes May FMP from 0.3287 to 0.3650. At .99, only four May merges remain, one adjudicable and wrong; it fails the volume rule. March's logistic .99 point makes no merges and has undefined decision FMP.

6.3 Merge support and pair-level changes

Decision FMP changes alongside adjudicable support (Table 3). On May, Band increases adjudicable-merge coverage from 28.73% to

Table 1: Primary event-time replay. Every row uses the same five scorer seeds $\{1, 2, 3, 4, 13\}$, a March-selected threshold of .0358, and a true capacity of 20. Precision/recall evaluate the known workflow domain; decision FMP and coverage use the merge denominators defined in Section 3. May is exploratory; July follows the newly frozen protocol.

Window	Policy	F_1 (mean±SD)	Precision	Recall	Decision FMP	Coverage
May	Arrival	0.415 ± 0.003	0.395	0.437	0.439	0.293
May	Band	0.449 ± 0.016	0.526	0.391	0.309	0.344
May	Gate	0.467 ± 0.014	0.510	0.431	0.340	0.309
May	Cutoff (10 d)	0.442 ± 0.003	0.473	0.416	0.378	0.320
May	Hidden band	0.415 ± 0.010	0.410	0.421	0.411	0.315
May	Hidden gate	0.425 ± 0.011	0.415	0.435	0.421	0.296
July	Arrival	0.469 ± 0.007	0.419	0.532	0.428	0.304
July	Band	0.488 ± 0.007	0.591	0.416	0.290	0.344
July	Gate	0.501 ± 0.007	0.551	0.460	0.340	0.311
July	Cutoff (10 d)	0.483 ± 0.006	0.514	0.457	0.358	0.323
July	Hidden band	0.473 ± 0.009	0.439	0.512	0.406	0.331
July	Hidden gate	0.475 ± 0.007	0.440	0.515	0.405	0.304

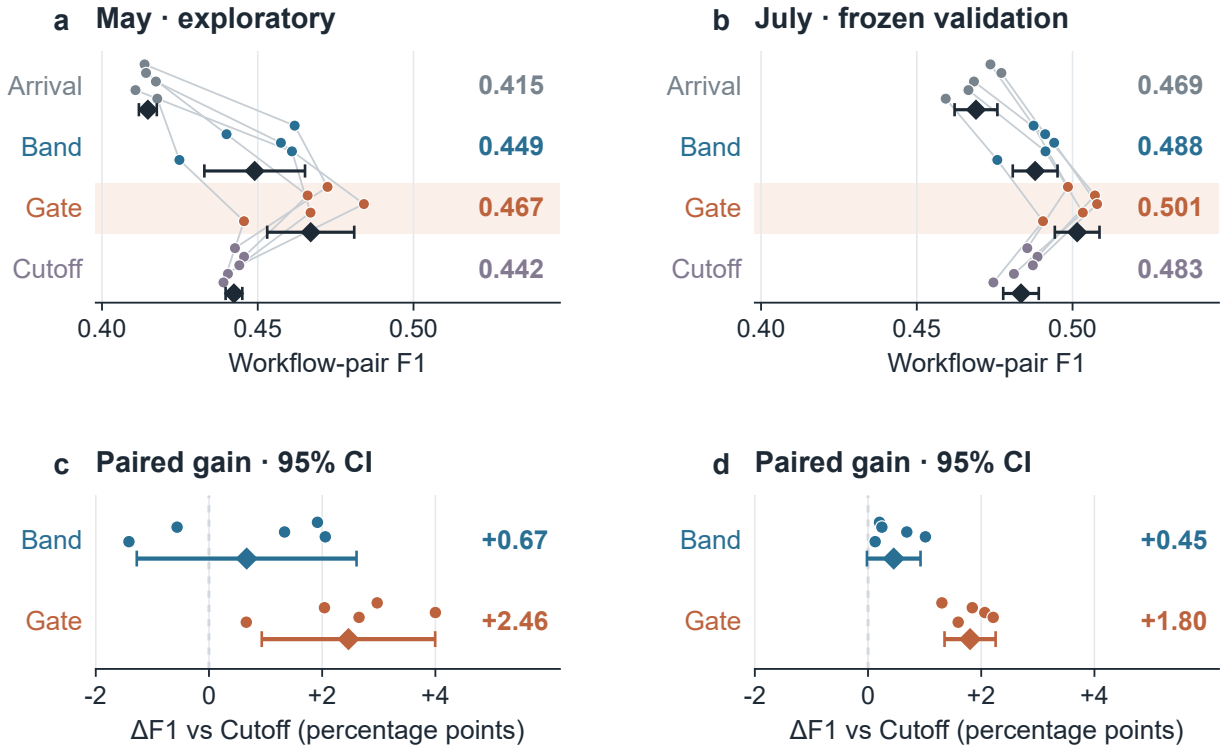


Figure 3: Matched performance and improvement beyond Cutoff. (a,b) All five GBT fits for each policy; thin lines connect the same fit. Black diamonds and whiskers show mean and sample SD. (c,d) All five matched differences from Cutoff; colored diamonds and whiskers show the mean and pointwise 95% t interval (four degrees of freedom, unadjusted). Differences are in F_1 percentage points; direct labels give means. Uncertainty describes fit-seed variability on fixed streams. Left: exploratory May. Right: newly frozen retrospective July validation.

32.74% while reducing the absolute adjudicable count from 2,391 to 2,248; Gate has 2,358. For common sources adjudicable under both policies, May FMP changes from 0.3911 to 0.3122 for Band ($n = 1,874$), and from 0.3929 to 0.3380 for Gate ($n = 2,092$). July gives 0.3907 to 0.2872 ($n = 1,894$) and 0.4034 to 0.3472 ($n = 2,169$), respectively. A pooled service-by-target-age standardization over

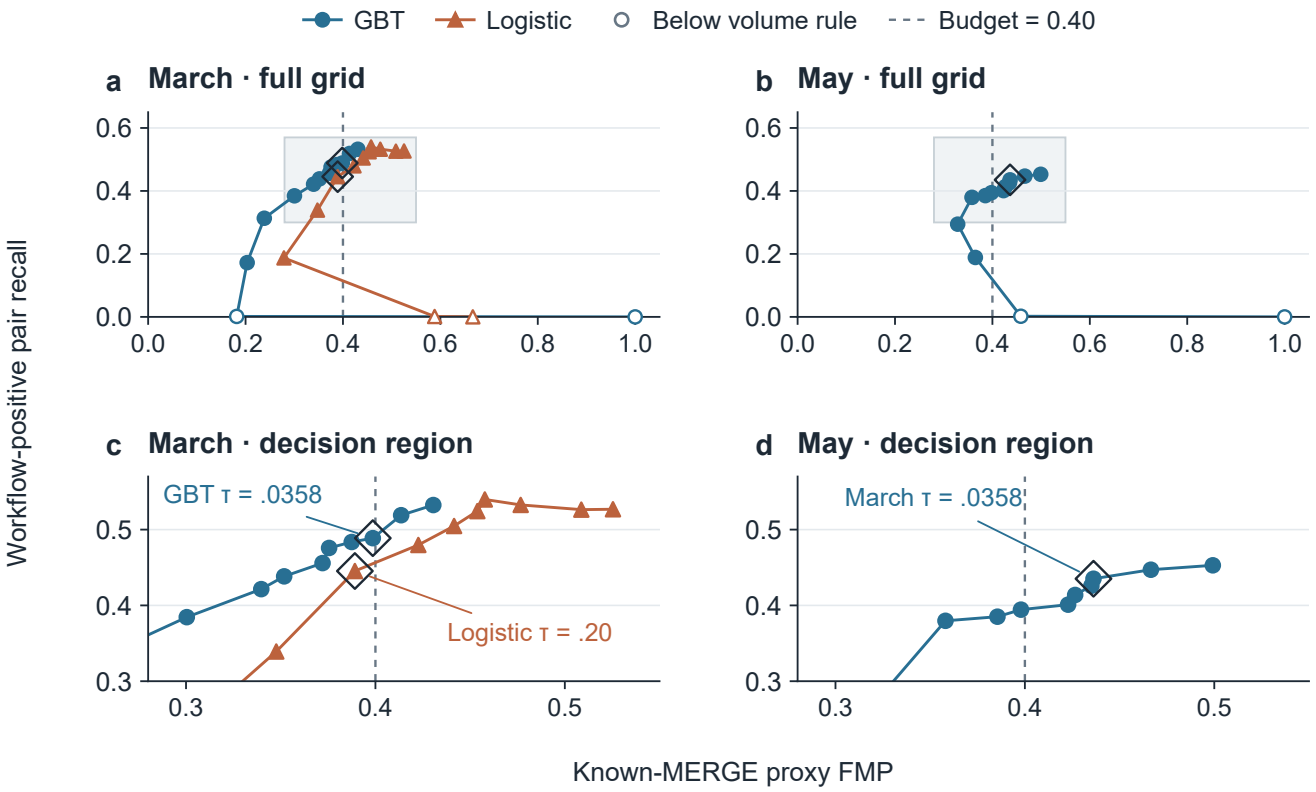


Figure 4: Complete arrival-policy threshold paths, seed 13. (a,b) Full grids; shaded rectangles locate the decision regions enlarged in (c,d). Filled points have at least 1% of requests merged and at least 100 known merges; open points fail this volume rule. Diamond outlines and direct labels identify the March-selected thresholds, including the fixed GBT threshold on May. Dashed lines mark the 0.40 known-merge proxy FMP budget. Straight connections follow threshold order. March logistic $\tau = .99$ has undefined risk, remains in Source Data, and is omitted from the plane.

Table 2: Logistic scorer, seed 13, separately March-calibrated $\tau = .20$. The gate and the same ten-day cutoff are both measured; these are single-fit descriptive comparisons.

Policy	May F_1	FMP	July F_1	FMP
Arrival	0.400	0.419	0.415	0.428
Band	0.299	0.328	0.313	0.314
Gate	0.432	0.366	0.411	0.383
Cutoff (10 d)	0.407	0.403	0.419	0.406

shared strata also retains negative May differences (−0.1152 and −0.0469). These comparisons condition on known support; truth for unknown merges remains unavailable, and selected targets can differ.

Both strategies recover and lose positive pairs (Figure 5). May Band gains 512 and loses 1,011, while Gate gains 506 and loses 673. July gains are 352/281 for Band/Gate, against losses of 1,095/742. Net positive-pair recall therefore falls for these runs. The gains in F_1 accompany a precision–recall tradeoff. The corresponding avoided/new wrong directed operations are 765/442 and 587/354 on May, and 874/419 and 626/361 on July. The artifact reports common, gained, lost, and neither-positive sets for every lag bin, with separate target-age bins for wrong operations.

The trigger audit separates claims at new arrivals from claims at other deadlines. May Band finalizes 1,316 pending targets by claim: 662 claims originate at a new arrival and 654 at another request's deadline. Gate's 611 claims split 542/69. Among Band's 512 newly recovered positive pairs, the later endpoint's operation is arrival-claim for 58, deadline-claim for 26, its own deadline for 50, and immediate output for 378. Gate's 506 split 36/2/14/454. These are associations among events in the realized execution; the causal mediation effect of new evidence remains unidentified.

6.4 Candidates: direct coverage and relaxed bounds

Actual scored query–representative pairs cover a fraction of reference positive pairs (Table 4). Partition recall can exceed this fraction through transitive attachment of unscored pairs. The union of actual representative edges forms an undirected graph. Within each connected component, a gold-aware construction of pure connected

Table 3: Complete merge denominators for seed 13. Each May arm finalizes all 35,915 requests and each July arm all 37,372, exactly once; NEW + MERGE equals the request count and final pending count is zero. Known merges = Correct + Wrong. Every outcome is relative to the closed-world workflow partition.

Window	Policy	NEW	MERGE	Correct	Wrong	Unknown	Coverage
May	Arrival	27,593	8,322	1,348	1,043	5,931	0.287
May	Band	29,048	6,867	1,528	720	4,619	0.327
May	Gate	28,100	7,815	1,548	810	5,457	0.302
May	Cutoff (10 d)	28,582	7,333	1,397	898	5,038	0.313
May	Hidden band	28,268	7,647	1,329	997	5,321	0.304
May	Hidden gate	27,686	8,229	1,377	1,014	5,838	0.291
July	Arrival	29,179	8,193	1,412	1,104	5,677	0.307
July	Band	30,674	6,698	1,559	649	4,490	0.330
July	Gate	29,734	7,638	1,538	839	5,261	0.311
July	Cutoff (10 d)	30,177	7,195	1,453	860	4,882	0.321
July	Hidden band	29,861	7,511	1,419	1,008	5,084	0.323
July	Hidden gate	29,297	8,075	1,449	1,022	5,604	0.306

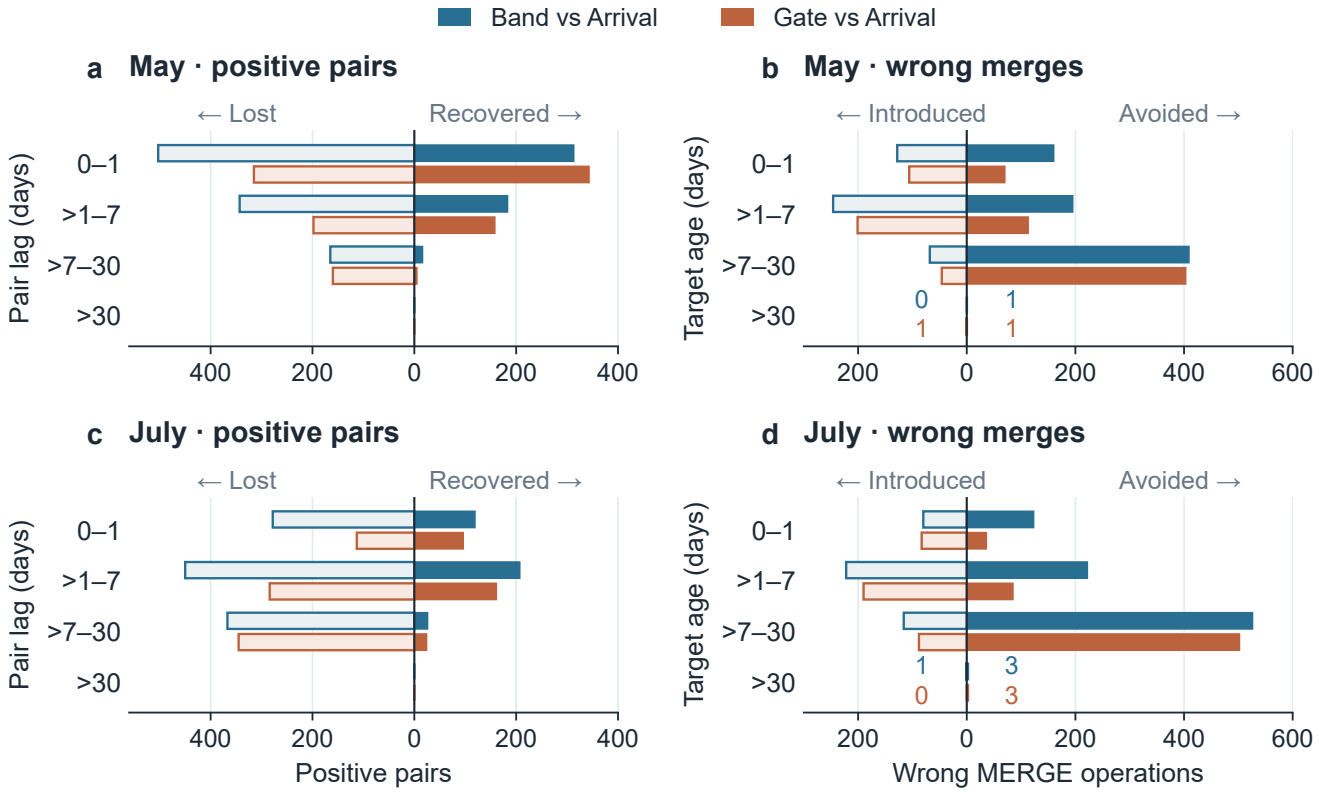


Figure 5: Exact set changes relative to Arrival (GBT seed 13). (a,c) Newly recovered and lost positive pairs, grouped by gold-pair lag. (b,d) Avoided and new directed wrong merges. Avoided operations use Arrival's target age; introduced operations use the deferral policy's target age. Rows show May and July. Bars extend left for losses or introduced errors, and right for recoveries or avoided errors; axis values are count magnitudes. Wrong-merge counts in the last age bin are labeled directly. Bins are $[0, 1]$, $(1, 7]$, $(7, 30]$, and $(30, \infty)$ days. Counts are dependent descriptive set differences under the workflow proxy. Unchanged pairs and trigger decompositions are included in Source Data.

Table 4: Candidate diagnostics, seed 13. All three measures are recall fractions (0–1). Direct coverage uses actually scored positive pairs. Static lower/upper bounds concern the best gold-aware partition of the realized representative graph with size cap 20; they relax online timing. The uncapped transitive upper bound is 1.0 throughout.

Window	Policy	Direct recall	Lower recall	Upper recall
May	Arrival	0.239	0.378	0.907
May	Band	0.338	0.526	0.907
May	Gate	0.290	0.499	0.907
May	Cutoff (10 d)	0.240	0.374	0.907
May	Hidden band	0.263	0.420	0.907
May	Hidden gate	0.248	0.396	0.907
July	Arrival	0.290	0.448	0.997
July	Band	0.429	0.628	0.997
July	Gate	0.352	0.530	0.997
July	Cutoff (10 d)	0.288	0.422	0.997
July	Hidden band	0.320	0.489	0.997
July	Hidden gate	0.303	0.465	0.997

groups of at most 20 yields a feasible static lower bound. A combinatorial upper bound distributes each gold label into groups of size 20 while relaxing connectivity within those groups. The fully uncapped reachability bound is 1.0 in both months. Both bounds are static and conditional on each policy's realized graph; they ignore arrival order and alternative retrieval histories.

Freezing Arrival's recorded candidate rows provides a separate diagnostic. Static threshold selection yields May $F_1 = 0.4177$; adding the ten-day age filter to those same rows gives 0.4003. This differs from the adaptive cutoff replay, which can change later representatives and candidates. The frozen analysis cannot implement pending states, later retrieval, or a waiting counterfactual. Historical external-linker runs use their own blocking and likewise cannot isolate commitment on a shared candidate population.

6.5 Actual delay and resource costs

Every core replay emits one final event per request and ends with zero pending requests. At $D = 7$, no held record exceeds seven elapsed days, including the end-of-month drain (Table 5). Hiding pending identities changes actual latency: all hidden records reach their own deadline, while visible targets can finalize on a claim. May's mean pending occupancy over the full replay horizon is 913.7 for Band and 251.9 for Gate, versus 1,145.2 and 475.2 for their hidden counterparts. This includes the final timer drain.

The one-fit May delay sensitivity gives Band F_1 of 0.4304 at $D = 3$ and 0.4203 at $D = 14$; Gate gives 0.4462 and 0.4437. The median, p95, and maximum held-request delays equal the chosen deadline in all four runs, with zero overruns. Longer permitted waiting is not uniformly beneficial. April's one-fit sensitivity is also nonmonotonic: band widths .3, .5, .7 give $F_1 = .4821, .4753, .4169$. Cutoffs 5/10/14 days give .4208, .4519, .4496. July's settings remain frozen; these grids are descriptive.

Table 6 reports the same trace-enabled May runs. The all-request mean covers 35,915 requests; held-request means cover 5,740 Band and 1,867 Gate requests ("–" denotes an empty held set). Peak pending counts requests. Wall and process CPU time include candidate-snapshot construction, serialization/writes, final close/flush, and the end-of-window timer drain; they exclude data/model loading, fitting, and post-replay metrics/output persistence. Trace time measures serialization and write calls, excluding the final close/flush. These timings characterize the six-worker shared CPU setting; isolated service capacity remains unmeasured.

Algorithm 2 Complete event-prefix comparison

Require: Stream S, cutoff T, fixed model/state/randomness ω
1: $S' \leftarrow$ preserve every arrival through T; perturb later content
2: $L \leftarrow \textsc{Replay}(S, \omega)$; $L' \leftarrow \textsc{Replay}(S', \omega)$
3: $P \leftarrow [e \in L : e.\text{time} \le T]$
4: $P' \leftarrow [e \in L' : e.\text{time} \le T]$
5: Assert $|P| = |P'|$
6: **for** $j = 1, \ldots, |P|$ **do**
7: Assert every field of P_j equals the corresponding field of P'_j
8: **end for**

7 OUTPUT-PREFIX AUDITING

For a fixed model, initial state, and randomness, we compare the full emitted event sequences of an original and a suffix-perturbed stream. At a real timestamp T, we independently filter both sequences by emission time $\le T$. Their length, order, time, request, action, target, and payload must agree. Truncating the perturbed run to the original prefix length would miss extra early emissions. The executable comparator checks extra, missing, and prematurely emitted events as well as altered ordering, targets, and randomness.

The real-stream check uses 8,153 May 1–7 requests and three cutoffs preserving 2,040, 4,077, and 6,118 arrivals, respectively. With the gate policy, $\tau = .0358$, and $D = 7$, all three normal output-prefix comparisons match. An intentionally invalid positive control uses the complete stream's time-weighted description length to alter actual pair scoring before replay. All three control runs change full output payloads. At the first two cuts, only score fields differ; the third changes one emitted decision. The check detects this injected future-information fault; other leakage pathways remain untested. The overlapping cutoffs provide three dependent diagnostic checks.

A targeted post-review check uses all 35,915 May requests, the same fixed gate scorer, and a cutoff of May 20 at 07:00 UTC (midnight local time). It preserves 21,683 arrivals and perturbs the subsequent 14,232. Before the cutoff, the 21,328 outputs comprise 20,961 arrival, 229 claim, and 138 deadline finalizations; claim counts refer to the pending target's NEW. All 138 own-deadline re-scorings occur at exactly seven days and before the month-end drain. The complete output prefixes match field for field after suffix perturbation. At 96 actual equal-time deadline/arrival timestamps, all 98 deadline calls precede arrivals. The instrumented baseline also matches the prior complete May event log. This extends observed deadline coverage; the scope of the causal guarantee is unchanged.

Two separate processes set `PYTHONHASHSEED` to 0 and 1, respectively, and replay the same complete week with the band policy and fixed scorer. Both emit 8,153 events and produce identical SHA-256 hashes of their complete serialized event logs. Stable keys, rather

Table 5: Actual finalization timing at $D = 7$ (seed 13); all latency columns are elapsed days. Claim counts the pending target finalized as NEW, whereas Deadline counts its own timer resolution. Held = Claim + Deadline; none exceeds seven days. A claim may be initiated by a fresh arrival or by another deadline; the operation-level decomposition is reported separately.

Window	Policy	Held	Claim	Deadline	All mean	Held mean	Held p95	Maximum
May	Band	5,740	1,316	4,424	0.97	6.04	7.00	7.00
May	Gate	1,867	611	1,256	0.27	5.12	7.00	7.00
May	Hidden band	6,208	0	6,208	1.21	7.00	7.00	7.00
May	Hidden gate	2,575	0	2,575	0.50	7.00	7.00	7.00
July	Band	5,670	1,350	4,320	0.91	5.99	7.00	7.00
July	Gate	1,869	619	1,250	0.26	5.11	7.00	7.00
July	Hidden band	6,131	0	6,131	1.15	7.00	7.00	7.00
July	Hidden gate	2,561	0	2,561	0.48	7.00	7.00	7.00

Table 6: Replay resources on May, GBT seed 13. Mean elapsed waiting is in days; wall, process CPU, and trace-write time are in seconds. All four arms use candidate tracing and the same six-worker CPU setup.

Policy	Mean wait (d)		Peak	Wall	CPU	Trace
	All	Held	pending	(s)	(s)	(s)
Arrival	0.00	–	0	100.12	99.08	6.49
Band	0.97	6.04	1,259	113.80	112.64	7.20
Gate	0.27	5.12	509	104.66	103.53	6.73
Cutoff	0.00	–	0	85.64	84.70	6.71

Table 7: Louisville's short-window check. Intervals are 2.5–97.5% quantiles of 4,000 issue-resampling draws, conditional on the selected candidate set; they are not independent-pair population confidence intervals.

Pair lag (days)	Same / eligible	Rate (%)	Interval (%)
0–1	43 / 537	8.01	3.88–13.64
2–3	18 / 366	4.92	1.56–8.02
4–6	35 / 398	8.79	2.01–16.78
7–13	38 / 639	5.95	1.10–10.58

than process object identity, determine ties. The audit configuration was fixed separately before the main comparisons; its boosted threshold coincides with the March-selected primary value. Full models, source versions, and configuration hashes are retained for reproduction.

8 TRANSFER AND ANNOTATION BOUNDARIES

Chicago. The historical Chicago check uses an earlier engine and a six-field export from the city's 311 portal [9], without free text. Among linked reports the median lag is 0.42 days and 84% fall within seven days. In its first March week, the largest observed band change among the tested settings is +0.0038 F_1; the selected setting yields a change in the following week's F_1 of −0.0078 in one fitted run. This descriptive, single-fit comparison leaves the repaired gate untested in Chicago; the observed changes do not establish a zero effect. Its code, inputs, and original results remain versioned separately from the corrected San Diego experiments.

Louisville annotations. The human-label check begins with 327 Louisville 311 records [24] across two weekly waves. The written instructions require separate A/B issue clustering, followed by a third adjudicator. They define an issue as a continuing problem or handling cycle on the same object. Matching category or address alone is insufficient; a new failure after a problem has been resolved should form a separate issue. Annotators are instructed to avoid the other annotator's decisions, model scores, and administrative identity information. Returned-file hashes match the recorded adjudication inputs and final labels. The hashes establish file consistency; compliance with the annotator-independence instructions remains unverified. Agreement measures must be interpreted in the context of the annotation task [4]; we report pre-adjudication agreement separately from the final labels.

A/B provisional clusterings agree on 90 positive pairs, disagree on 286, and agree on 52,925 negative pairs. Positive-pair Dice agreement is 0.386 and adjusted Rand index [18] is 0.384. Excluding any record flagged for adjudication leaves 258 records, with Dice 0.487 and adjusted Rand index 0.486; this sensitivity estimate is conditional on a selected-confidence subset and does not replace the full-sample agreement measures. Adjudication retains 292 records in 226 issues, including 26 multi-record issues, and leaves 35 unadjudicable. All 326 asserted same/different review judgments agree with the resulting partition. Unadjudicable records are excluded from the lag analysis rather than treated as singleton negatives.

Among same-service pairs passing the recorded token-overlap rule, the four lag-bin same-issue rates are nonmonotonic (Table 7). We preserve within-issue dependence by resampling final issues with replacement. A within-issue pair receives its issue's multiplicity; a between-issue pair receives the product of the two multiplicities. The resulting sensitivity intervals are broad. The data comprise two waves, and all 639 pairs in the last bin cross waves, leaving residual wave and service dependence. The sample spans 13 days. These rates leave flatness unresolved and do not directly test a mechanism extending through 8–30 days. The released anonymous cluster and candidate-pair relationships reproduce the agreement and resampling calculations conditional on the recorded candidate set.

9 DISCUSSION AND VALIDITY

What is identified. Visibility changes retrieval, claim opportunities, and the state encountered by later arrivals. Realized waiting depends on claim opportunities under a shared nominal deadline. A net recall decrease or an interval containing zero is compatible with later evidence helping individual records. The residual over a cutoff is a complete policy contrast, not an additive decomposition of retrieval and waiting.

Reference labels and support. Workflow labels represent administrative linkage practices, and cross-component matches can be missing. Decision FMP is conditional on the records a policy merges. Denominator and common-support analyses summarize this selected set; unknown labels and nonrandom support remain. Closed-request sampling can omit unresolved cases, and label-creation times in the annual export are unknown.

Scope of generalization. Five fits quantify fitting variability on fixed streams. July provides retrospective temporal validation after earlier policy exploration. The logistic check has one fit; Chicago is historical; Louisville is selected and adjudication-dependent. Multi-city evaluation and label-acquisition histories would be needed for stronger operational claims.

Candidate and tool comparisons. Frozen-trace filtering and graph bounds characterize recorded candidates; adaptive replay captures policy feedback. The historical Splink runs provide end-to-end comparisons with their own blocking and estimation. Their original environment was not recorded. Our API check used Splink 5.0.0, which supports term frequency adjustments; the old configurations left them disabled. Commitment, blocking, and estimation differ jointly across these systems.

Computation and reproducibility. Deferred output has a latency cost under the exact timer schedule. The reported times come from six concurrent CPU workers and include candidate-snapshot serialization in traced runs. These measurements describe instrumented concurrent replay performance.

10 ARTIFACT AND REPRODUCIBILITY

The public repository is https://github.com/yyting2006-ai/deferral-streaming-linkage. Release v0.2.0 contains the corrected engine, complete policy and calibration configuration, individual result rows, paired analysis, candidate-bound code, event-prefix comparator, tests, figure sources, and checksums. The released San Diego snapshot and frozen scorer recipe allow raw event logs to be regenerated. The snapshot's checksum is checked before replay. The Chicago snapshot supports the separately versioned historical check. The Louisville derivative contains fresh anonymous identifiers, cluster relationships, pending flags, wave membership, and precomputed eligible-pair lag bins. Its reproducible scope is agreement and issue-resampling from the recorded relationships; the original text-based eligibility screening and annotation conduct remain outside that scope. Release v0.2.2 adds the post-deadline audit, resource summary, and current manuscript figure mapping.

The earlier engine and its results remain under v0.1.0. The reproduction instructions distinguish the versions and require a fresh output directory. Cache reuse is rejected when configurations or provenance differ. Undefined metrics retain their undefined status in the analysis. Preparation, calibration selection, and run schedules are recorded explicitly, including any unknown original selection times.

11 CONCLUSION

Retrievable provisional identities improve boosted-scorer F_1 on the two measured streams under workflow-derived labels, and the staleness gate exceeds a ten-day cutoff in matched fits. Higher aggregate scores coexist with fewer wrong merges and with both recovered and lost positive pairs. Visibility changes actual waiting and future candidates together. The July logistic reversal shows that the gate–cutoff ordering depends on the scorer. The explicit event-time contract and complete output audit make the policy comparisons reproducible and record their latency and support costs.

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